CURRICULUM VITAE

May, 2026

PERSONAL:

Name: Dmitrii Panasenko

Birth Date: October 5, 1995

Birth Place: Chelyabinsk, Russia

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Google Scholar: `https://scholar.google.com/citations?user=v6wsLukAAAAJ`

EDUCATION:

2023–present: PhD Program, Department of Mathematics and Mathematical Statistics, Umeå University, Umeå, Sweden. Advisor: Gerold Jäger.

2019–2023: Postgraduate Program, Department of Algebra and Topology, Krasovskii Institute of Mathematics and Mechanics, Yekaterinburg, Russia. Advisor: Vladislav V. Kabanov.

2013–2019: Specialist Program, Department of Mathematics, Chelyabinsk State University, Chelyabinsk, Russia. Primary course: mathematics, computer science. Specialization: computer security. Graduation thesis: "A new infinite series of Neumaier graphs as graphs in which cliques induce completely regular codes" (revised into paper "The smallest strictly Neumaier graph and its generalisations"). Advisor: Sergey V. Goryainov.

EMPLOYMENT:

2023–present: PhD Student, Department of Mathematics and Mathematical Statistics, Umeå University, Umeå, Sweden. Advisor: Gerold Jäger.
Courses taught: Linear Programming (teaching assistance), Integer Programming (problem solving), Discrete Modeling (problem solving), Computer Security (lectures), Reverse Engineering (lectures), Methods and Tools for Computer Scientists (problem solving).

2019–2022: Senior Lecturer, Department of Mathematics, Chelyabinsk State University, Chelyabinsk, Russia.
Courses taught: Finite Fields (problem solving), Graph Theory and its Applications (problem solving), Theory of Automata and Formal Languages (problem solving), Assembly Languages (lectures and problem solving), Number Theory Methods in Cryptography (lectures and problem solving), Security Models of Computer Systems (lectures and problem solving), Information Theory (lectures and problem solving), Mathematical Logic and Theory of Algorithms (problem solving), Introduction to Computer Security (lectures and problem solving).

2019–2022: Laboratory Assistant, Laboratory of Technical Information Security, Chelyabinsk State University, Chelyabinsk, Russia.

RESEARCH INTERESTS:

Traveling Salesman Problem, Minimal Spanning Tree Problem, single and set tolerances, strongly regular graphs, Deza graphs, divisible design graphs, Neumaier graphs, equitable partitions of graphs, eigenfunctions of graphs, enumeration of graphs.

MAINTAINED WEBSITES:

`http://alg.imm.uran.ru/dezagraphs/main.html`

SKILLS:

SageMath, Python, C, C++, Assembly x86/x64, Magma.

GRANTS:

- 2022–2023, Russian Science Foundation, "Extremal eigenfunctions of distance-regular graphs", **participant**, Chelyabinsk State University.
- 2020–2021, Joint project of Russian Foundation for Basic Research and National Natural Science Foundation of China, "Extremal problems in spectral graph theory", **participant**, Chelyabinsk State University.
- 2017–2019, Russian Foundation for Basic Research, "Algebraic and combinatorial methods in graph theory", **participant**, Sobolev Institute of Mathematics.
- 2016–2017, Russian Foundation for Basic Research, "Vertex-transitive Deza graphs", **participant**, Krasovskii Institute of Mathematics and Mechanics.

PUBLICATION LIST:

1. S. Goryainov, D. Panasenko, *On eigenfunctions of the block graphs of geometric Steiner systems*, Journal of Combinatorial Designs, 32(11) (2024), pages 629–641, `https://doi.org/10.1002/jcd.21951`
2. D. Panasenko, *The vertex connectivity of some classes of divisible design graphs*, Siberian Electronic Mathematical Reports, 19(2) (2022), pages 426–438, `http://semr.math.nsc.ru/v19/n2/p426-438.pdf`
3. D. Panasenko, L. Shalaginov, *Classification of divisible design graphs with at most 39 vertices*, Journal of Combinatorial Designs, 30(4) (2022), pages 205–219, `https://doi.org/10.1002/jcd.21818`
4. S. Goryainov, D. Panasenko, L. Shalaginov, *Enumeration of strictly Deza graphs with at most 21 vertices*, Siberian Electronic Mathematical Reports, 18(2) (2021), pages 1423–1432, `https://doi.org/10.33048/semi.2021.18.107`
5. R. J. Evans, S. Goryainov, D. Panasenko, *The smallest strictly Neumaier graph and its generalisations*, The Electronic Journal of Combinatorics, 26(2) (2019), #P2.29. `https://doi.org/10.37236/8189`
6. S. Goryainov, D. Panasenko, *On vertex connectivity of Deza graphs with parameters of complements to Seidel graphs*, European Journal of Combinatorics, 80 (2019), pages 143-150. `https://doi.org/10.1016/j.ejc.2018.10.006`

CONFERENCE/SEMINAR PARTICIPATION:

- *Computing regular set tolerances for combinatorial minimization problems* / The Swedish Mathematical Society Autumn Meeting 2025 in Uppsala University, Uppsala (Sweden), 2025, `https://www.swe-math-soc.se/pdf/program-host2025.pdf`.
- *Deza graphs and vertex connectivity* / Lunch Seminar in Combinatorial Optimization in Eindhoven University of Technology, Eindhoven (Netherlands), 2024, `https://spor.win.tue.nl/calendar/category/co/co-seminar/list/?eventDisplay=past`.
- *Deza graphs and vertex connectivity* / Algebraic Graph Theory Seminar in University of Waterloo, Waterloo (Canada), 2024, `https://math.uwaterloo.ca/~agtheory/`.
- *Deza graphs and vertex connectivity* / Discrete Mathematics Seminar in Umeå University, Umeå (Sweden), 2024, `https://www.umu.se/en/research/groups/discrete-mathematics/seminars/`.
- *On eigenfunctions of block graphs of Steiner triple systems* / joint with S. Goryainov / The 7th Workshop on Algebraic Graph Theory and its Applications, Novosibirsk (Russia), 2022, `https://mca.nsu.ru/agt7/`.
- *More about vertex connectivity of special classes of divisible design graphs*, The 6th Workshop on Algebraic Graph Theory and its Applications, Novosibirsk (Russia), 2022, `https://mca.nsu.ru/agt6/`.
- *Vertex connectivity of some classes of divisible design graphs*, The 5th Workshop on Algebraic Graph Theory and its Applications, Novosibirsk (Russia), 2021, `https://mca.nsu.ru/atg5_2021/`.
- *Enumeration of divisible design graphs* / joint with L. Shalaginov / The International Conference and PhD-Master Summer School "Groups and Graphs, Semigroups and Synchronization", Sochi (Russia), 2021, `http://conf.uran.ru/Default?cid=g2s2`.
- *Enumeration of divisible design graphs* / joint with L. Shalaginov / The 4th Workshop on Algebraic Graph Theory and its Applications, Novosibirsk (Russia), 2021, `https://mca.nsu.ru/atg4_2021/`.
- *On computation of WL-closures of small Deza graphs*, The 3rd Workshop on Algebraic Graph Theory and its Applications, Novosibirsk (Russia), 2020, `https://mca.nsu.ru/atg3_2020/`.
- *The complete enumeration of strictly Deza graphs with at most 21 vertice* / joint with L. Shalaginov / The 2nd Workshop on Algebraic Graph Theory and its Applications, Novosibirsk (Russia), 2020, `http://mca.nsu.ru/atg2020/`.
- *The complete enumeration of strictly Deza graphs with at most 21 vertices* / joint with L. Shalaginov / The International Conference and PhD-Master Summer School "Groups and Graphs, Designs and Dynamics", Yichang (China), 2019, `https://math.sjtu.edu.cn/conference/G2D2/`.
- *On equitable partitions of divisible design graphs* / joint with S. Goryainov and L. Shalaginov / The International Conference and PhD-Master Summer School "Groups and Graphs, Representations and Relations", Novosibirsk (Russia), 2018, `http://math.nsc.ru/conference/g2/g2r2/`.
- *Cycles in the 3-Big Pancake graphs* / joint with S. Goryainov, E. Konstantinova and A. Pankratova / The International Conference and PhD-Master Summer School "Groups and Graphs, Metrics and Manifolds", Yekaterinburg (Russia), 2017, `https://g2.imm.uran.ru/g2m2/`.